

Determinants

ANSWER KEY



2x2 determinants

- 1 Find $\det \begin{bmatrix} 4 & 3 \\ 2 & 5 \end{bmatrix}$.

$$4(5) - 3(2) = 20 - 6 = 14.$$

3x3 determinants via cofactor expansion

- 2 Find $\det \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix}$ by cofactor expansion along the first row.

$$1 \det \begin{bmatrix} 1 & 4 \\ 6 & 0 \end{bmatrix} - 2 \det \begin{bmatrix} 0 & 4 \\ 5 & 0 \end{bmatrix} + 3 \det \begin{bmatrix} 0 & 1 \\ 5 & 6 \end{bmatrix} = 1(-24) - 2(-20) + 3(-5) = -24 + 40 - 15 = 1.$$

The determinant product rule

- 3 For $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$, verify $\det(AB) = \det(A) \det(B)$.

$$\det A = 4 - 6 = -2, \det B = 6 - 0 = 6, \text{ so } \det A \det B = -12. AB = \begin{bmatrix} 2 & 7 \\ 6 & 15 \end{bmatrix}, \text{ and}$$

$$\det(AB) = 2(15) - 7(6) = 30 - 42 = -12. \text{ They match.}$$

The invertibility criterion

- 4 State the invertibility criterion in terms of the determinant.

A square matrix A is invertible if and only if $\det A \neq 0$.

- 5 Is $\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$ invertible? Justify using the determinant.

$$\det = 2(2) - 4(1) = 0. \text{ Since the determinant is 0, the matrix is NOT invertible (singular).}$$

Applying the invertibility criterion

- 6 For what value(s) of k is $\begin{bmatrix} k & 2 \\ 3 & k \end{bmatrix}$ NOT invertible?

$$\det = k^2 - 6. \text{ Setting } k^2 - 6 = 0 \text{ gives } k = \pm\sqrt{6}. \text{ The matrix fails to be invertible exactly when } k = \sqrt{6} \text{ or } k = -\sqrt{6}.$$

More 2x2 determinant practice

7 Find $\det \begin{bmatrix} 7 & 2 \\ 3 & 5 \end{bmatrix}$.

$$7(5) - 2(3) = 35 - 6 = 29.$$

More 3x3 cofactor expansion practice

8 Find $\det \begin{bmatrix} 2 & 0 & 1 \\ 3 & 1 & 2 \\ 1 & 0 & 4 \end{bmatrix}$ by expansion along the second column (it has two zeros).

Only $a_{22} = 1$ is nonzero in column 2. $\det = 1 \cdot \det \begin{bmatrix} 2 & 1 \\ 1 & 4 \end{bmatrix} = 1(8 - 1) = 7$. (Cross-check by expanding along

row 1: $2 \det \begin{bmatrix} 1 & 2 \\ 0 & 4 \end{bmatrix} - 0 + 1 \det \begin{bmatrix} 3 & 1 \\ 1 & 0 \end{bmatrix} = 2(4) + 1(-1) = 7$. ✓)

Determinants of triangular matrices

- 9 State the shortcut for the determinant of a triangular matrix (all entries above, or below, the main diagonal are zero).

The determinant equals the product of the diagonal entries.

10 Use this shortcut to find $\det \begin{bmatrix} 3 & 0 & 0 \\ 5 & -2 & 0 \\ 1 & 7 & 4 \end{bmatrix}$.

$$\text{Product of the diagonal: } 3 \times (-2) \times 4 = -24.$$

Determinants and row operations

- 11 If you swap two rows of a matrix, how does the determinant change?

Its sign flips (multiplied by -1); the magnitude is unchanged.

- 12 If a matrix has two identical rows, what must its determinant be, and why?

Swapping the two identical rows leaves the matrix unchanged but must flip the determinant's sign — the only number equal to its own negative is 0, so the determinant must be 0.

Application: area of a parallelogram

- 13** The area of the parallelogram spanned by $\mathbf{u} = \langle a, b \rangle$ and $\mathbf{v} = \langle c, d \rangle$ equals $\left| \det \begin{bmatrix} a & b \\ c & d \end{bmatrix} \right|$. Find the area for $\mathbf{u} = \langle 3, 1 \rangle$, $\mathbf{v} = \langle 1, 4 \rangle$.

$$\det \begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix} = 12 - 1 = 11. \text{ Area} = |11| = 11.$$

Scalar multiples and the determinant

- 14** For an $n \times n$ matrix A , $\det(cA) = c^n \det(A)$. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = 2A$, find $\det(B)$ using this rule, then verify directly.

$$\det A = 4 - 6 = -2. \text{ Rule: } \det(B) = 2^2 \det(A) = 4(-2) = -8. \text{ Direct check: } B = \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix},$$

$$\det B = 16 - 24 = -8. \checkmark$$

The determinant of the transpose

- 15** Verify that $\det(A^T) = \det(A)$ for $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

$$\det A = 4 - 6 = -2. A^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}, \det(A^T) = 4 - 6 = -2. \text{ Equal, as expected.}$$

The determinant of the inverse

- 16** For $A = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$, verify that $\det(A^{-1}) = \frac{1}{\det A}$.

$$\det A = 3(1) - 1(2) = 1. A^{-1} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix}, \det(A^{-1}) = 3 - 2 = 1. \text{ Since } \frac{1}{\det A} = \frac{1}{1} = 1, \text{ this matches}$$

$$\det(A^{-1}) = 1.$$

A quick mental-math determinant

- 17** Find $\det \begin{bmatrix} -2 & 5 \\ 0 & 3 \end{bmatrix}$ (notice it's already triangular).

$$\text{Triangular, so the determinant is the product of the diagonal: } (-2)(3) = -6.$$

- 18** Find $\det \begin{bmatrix} 6 & 2 \\ 3 & 1 \end{bmatrix}$, and note what it tells you about invertibility.

$6(1) - 2(3) = 6 - 6 = 0$. The determinant is 0, so this matrix is singular (not invertible) — its rows are proportional ($\langle 6, 2 \rangle = 2\langle 3, 1 \rangle$).